

INDRA: Interactive Deep-dreaming Robotic Artist. Painting with a Real-Time Embodied AI

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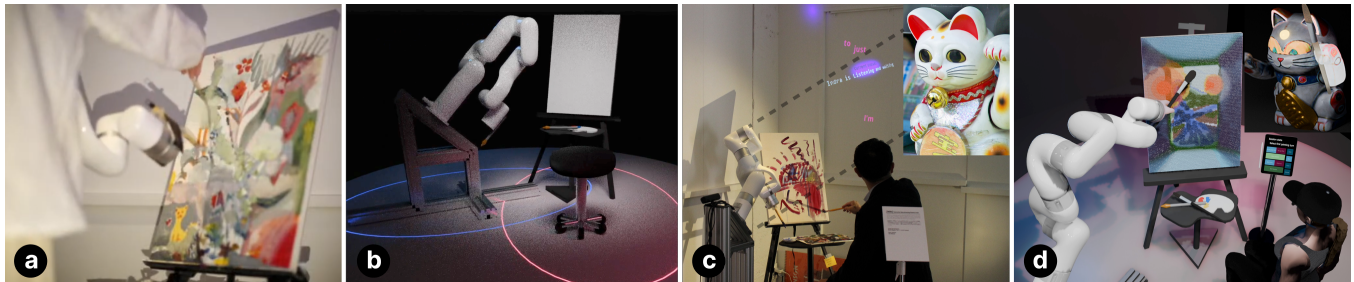


Figure 1: (a) INDRA was exhibited over three days where participants engaged in a collaborative, turn-based painting session with a robotic arm. (b) The two entities are represented in zones. (c) The robot arm motions are synchronized with the visual feedback of the anthropomorphized AI agent on a display. (d) Turn switching and level control is through a touch tablet.

Abstract

We present INDRA, an Interactive Deep-Dreaming Robotic Artist, designed to assist in real-time, embodied collaboration on a traditional painting canvas. INDRA enhances fine art creation by dynamically adapting to users' needs, offering support throughout the artistic process. Visually represented as a cybernetic avatar of a mechanical Maneki-neko, the Japanese beckoning cat, INDRA serves as a creative AI companion that paints alongside artists. Its mission is to improve artist-friendly perception of AI among creatives, blending technology with traditional arts to inspire innovation. INDRA integrates robotic painting with advanced AI processes, including speech recognition, computer vision, and generative image synthesis to facilitate co-creation. INDRA differs from existing AI art generators and art robots in its turn-based framework, where user and robot actively share the same canvas.

CCS Concepts

• **Human-centered computing** → **Interaction paradigms**; • **Applied computing** → **Arts and humanities**.

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Keywords

Human-AI Collaboration; Robotic Painting; Generative AI; Embodied Interaction; Artistic Agency

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1 Introduction

Traditional artistic practices, including manual form painting and sculpture, remain fundamental to cultural expression. As artificial intelligence (AI) intersects with these disciplines, it is essential to consider human value-centered design, particularly when engaging traditional artists[8]. While AI-generated art has been widely explored, existing approaches often lack real-time, tangible collaboration. We introduce INDRA, a collaborative painting robot that takes sequential turns with human creators on a single canvas. This work examines how *human agency* evolves when integrating physical generative AI into a traditional art workflow. However, the language and design surrounding AI often imply an entity with its own intent, potentially undermining artists' sense of ownership and originality [7, 23]. Our goal is to *improve perceptions of generative AI among traditional artists* while preserving their *artistic agency*,

the ability to recognize themselves as the primary author of an artwork. Performing as a cybernetic avatar, INDRA not only produces a physical painting, but also incorporates the human artist's intent into the generative process. We measure artistic agency with the Sense of Agency Scale (SoAS), situating our work within a broader discourse on generative AI as a co-creator in visual arts. Our focus is on balancing the benefits of AI assistance with the need for artists to feel in control, thus shaping the future role of human creators in an era of computational creativity. The rise of AI-generated art has significant psychological and social implications for traditional artists. The increasing reliance has led to Creative Displacement Anxiety (CDA), a psychological response to AI-generated art. CDA highlights how AI's ability to mimic or surpass human creativity leads to identity loss, economic anxiety, and reduced artistic motivation among creatives. It also causes skepticism, alienation, and choice paralysis among consumers, emphasizing the need for ethical AI integration to mitigate its negative effects [5]. Socially, the online art community might experience a divide between proponents of AI art and those who reject its use due to misalignment of core values, suggesting the design of AI art technologies should be human value-centered[2]. Applying generative AI to traditional painting requires considering the effects on the existing painting community. Art Workers Japan found from a survey of creative field workers (25,560 responses) that 59% of creatives worry about losing work opportunities to AI, and 92% feared their work would be scrapped as art theft [22]. This highlights the concerns regarding AI potentially replacing human artists or misappropriating their creations.

Our Approach and Contributions To address these challenges, we propose INDRA, a real-time, user-aware AI agent that combines various generative AI techniques, including speech recognition, object detection, diffusion-based image synthesis, and robotic motion planning. This integrated system is designed for collaborative painting, producing work that is solely responsive to the user. By fostering this approach, INDRA encourages collaboration rather than competition. In this paper, we:

- **Introduce INDRA's layered architecture and real-time pipeline**, demonstrating how our approach balances user input with AI-driven creativity while maintaining a shared physical canvas.
- **Develop a method for dynamically interpreting multimodal cues** (verbal, visual, gestural) so INDRA can adapt its painting strategy on the fly based on user feedback.
- **Present example applications** in various visual art contexts, including accessibility-focused painting and educational scenarios, illustrating how INDRA fosters co-creative experiences and enhances artistic agency.

2 Related Work

Generative AI art tools like DALL-E [18], MidJourney, and Stable Diffusion[19] create digital images from text prompts, while robotic painting systems like Ai-Da and AICAN[1, 16] demonstrate how autonomous agents can physically produce paintings, yet lack interactive, *real-time* collaboration. CoFrida[20] moves toward interactivity but primarily involves a one-time pass from user input to AI-generated content, focusing on simulation rather than live,

turn-based painting. On the other hand, there is growing interest in “embodied AI”, where a system perceives and acts in its environment through a robotic body, allowing adaptive and tangible exchange[11, 13]. Embodied collaboration can foster more intuitive engagement than purely screen-based interactions [7, 11]. However, it raises design questions about how to separate control and responsibility for creative outputs. More often research on human-computer collaboration indicates that individuals prefer to maintain final authority over generative outputs[23]. Confusion can occur when AI decisions are unclear or unexpected. To address this, we structured turn-taking with a turn switch on a control tablet panel, which helps clarify control boundaries and may strengthen artists' sense of authorship. We measure this using the Sense of Agency Scale (SoAS) [21] in a pilot study.

In summary, current AI-driven art tools and robotic painting systems face several key limitations: they are often confined to digital formats, lack real-time responsiveness, and offer minimal turn-based collaboration. To address these gaps, we propose INDRA, a real-time, user-aware AI agent that merges speech recognition, computer vision, and diffusion-based image synthesis into a single interactive and adaptive system. Our design builds on insights from layered AI architectures [4, 9], which decompose complex tasks into modular levels of perception and control. Early research on subsumption [4] and layered agents [3, 9] illustrated how real-time reactivity could be achieved by distributing responsibilities across multiple system layers. Recent developments in these principles integrate high-level generative models, allowing for rapid adaptation to user interactions and changing environments[14]. INDRA leverages this layered design to combine diffusion models with ambient user inputs—spoken prompts, on-canvas sketches, and object recognition. A layered AI approach is particularly effective for *co-creative* painting, as each generative process addresses a distinct aspect of user interaction—visual, linguistic, or gestural—and collectively produces an evolving shared artwork [10, 12]. Unlike purely digital AI art applications, INDRA physically interacts with a shared canvas, creating a tangible and evolving collaboration. This tactile engagement introduces a layer of complexity and intimacy, prompting artists to explore the boundaries between control and creative partnership.

3 System

INDRA operates across varying levels of AI understanding, dynamically adapting to the context provided by the artist. To further enhance INDRA's approachability and clarify its internal processes, we developed a 3D avatar in the form of a mecha maneki-neko. The avatar's modules are separated into distinct body parts to provide real-time transparency into INDRA's internal processes, helping users understand how their voice, sketches, or physical painting are driving AI behavior.

- (1) **Embodiment and Approachability:** By adopting a playful, non-human form, INDRA leverages findings that anthropomorphic designs enhance user engagement and acceptance of AI systems [7, 15]. The mecha maneki-neko design draws inspiration from the Japanese cultural symbolism of a beckoning paw, which in this case symbolizes a robotic arm. These figures gesture towards the user, inviting them to

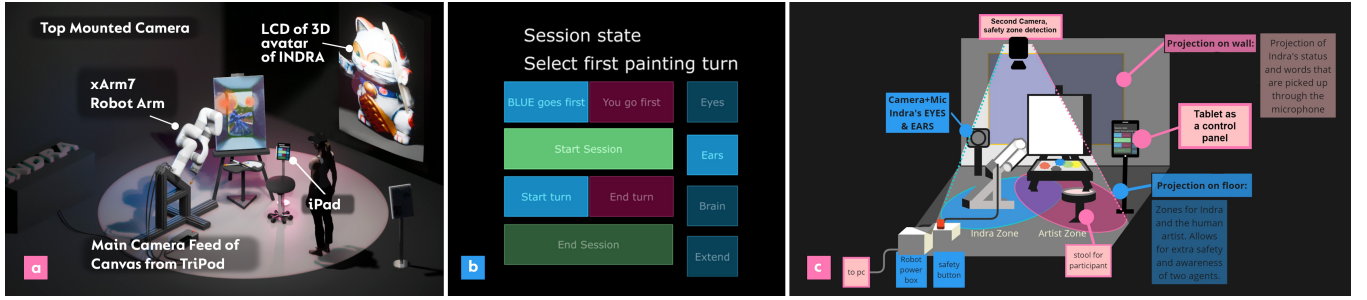


Figure 2: Diagram of system space layout (a) in a demo setting. Screenshot of touch tablet control panel display (b). A detailed map of hardware illustrate INDRA's perception sensors.(c)

paint together and reinforcing the system's role as a "friendly helper." This avatar is rendered on a nearby screen, visually mirroring the robotic arm's actions and indicating which AI modules are active.

- (2) **Synchronized Movement:** When the xArm is active, the avatar's arm moves in real time. This synchronization helps users form a mental model of INDRA as a single, coherent entity rather than disconnected hardware and software. We map xArm joint angles to the 3D model's arm for smooth motion correspondence.
- (3) **Visual Feedback of AI State:** The avatar's key features light up or animate to reflect which subsystems of INDRA (Ears, Eyes, Brain, Arm) are currently engaged:
 - **Eyes:** Brighten or change color when image processing is running. Utilizes visual abstraction of the canvas and stream diffusion from raw visual input with subtle augmentations.
 - **Ears:** Illuminate when INDRA's speech recognition is actively listening. Processes auditory input through speech recognition. OpenAI's Whisper API for multilingual speech recognition to transcribe and interpret verbal cues in English and Japanese. These spoken inputs influence INDRA's visual responses by affecting the prompt data set.
 - **Brain:** A glowing element in the avatar's head pulses during generative steps. Incorporates context from Yolov8, PaintsUNDO, and intention estimation. Employing computer vision for content recognition, allowing it to match context—such as adding an apple to a user's tree painting. This level also leverages PaintsUNDO AI, which separates INDRA's imagined painting from StreamDiffusion into frames that are trained on time-lapse drawing processes.
 - **Arm:** Highlights when the robotic arm is ready for manual override or responding to tablet commands. Executes commands and drawing actions from the tablet interface.

3.1 Software and Hardware

A core requirement for INDRA is real-time responsiveness, allowing users to engage with it as an active co-creator. To achieve this, we incorporate various modules including speech recognition, vision-based detection, stable diffusion, and robotic control through TouchDesigner, a visual programming environment ideal for managing asynchronous data streams, GPU-accelerated image

processing, and external device control [6]. In contrast to the Robot Operating System (ROS), which can delay execution due to blocking functions, TouchDesigner effectively synchronizes sensor inputs, AI pipelines, and robotic motion for dynamic interaction. The key components include Stable(Stream)Diffusion for image synthesis, OpenAI Whisper for speech recognition, YOLOv8 for object detection, and algorithms to convert digital sketches from an iPad into robotic brushstrokes. The system uses a uFactory xArm7 robotic arm mounted at a 45° angle for optimal motion, paired with a Canon EOS and web cameras for real-time canvas analysis and safety monitoring. A large LCD displays a 3D avatar indicating active modes, while a microphone captures ambient speech to enhance painting intentions, such as integrating floral motifs upon detecting references to "flowers."

4 User Experience

INDRA offers a versatile and inclusive user experience, dynamically adapting its behavior through sensor data and real-time user inputs. This includes visual feedback, voice commands, and physical interactions with the canvas. The tablet interface provides intuitive controls, featuring module buttons and a drawing window for an additional control layer, enabling artists to directly take command of the robot arm. With its segmented design, INDRA accommodates a wide range of users, enhancing the creative process for trained and novice artists. For experienced artists, INDRA provides dynamic feedback and real-time composition analysis, introducing elements that inspire innovation and expand artistic possibilities. Digital artists with limited traditional painting experience can bridge the gap to conventional media, using familiar digital tools to create physical works. For children, INDRA transforms painting into a playful, educational experience, teaching concepts like color mixing and composition through interactive collaboration. Meanwhile, for artists with physical disabilities, INDRA enables accessibility by translating digital tablet strokes into physical canvas movements, extending their creative reach and overcoming physical limitations. By customizing its features to suit a variety of user needs, INDRA effortlessly integrates into diverse creative workflows, fostering a unique and engaging artistic journey. At CHI Interactivity, participants will engage with INDRA for 5-7 minutes at a painting station featuring a canvas, robotic arm, and tablet interface. Wearing noise-canceling headphones, they will guide INDRA using ambient speech, their brush strokes, and tablet inputs. A LCD display shows



Figure 3: Child uses digital prototype of INDRA (left). Scenario usage of INDRA for physical disabled artists (middle). Application of INDRA in a large art studio as a cutting-edge AI tool for professional painters (right).



Figure 4: TouchDesigner system(left). Artist creating work (middle). INDRA and artist drawing(right).

INDRA's synchronized 3D model with the enabled components as body parts lit up. Thought process are overlaid on the screen, revealing keywords and AI-generated interpretations that guide its artistic contributions.

4.1 Methodology

To explore these dynamics, we use a mixed-method approach, that combines qualitative and quantitative data. Qualitative data is gathered through participatory observations, interviews, and surveys conducted with INDRA users during exhibition. For the quantitative aspect, we use the Sense of Agency Scale (SoAS) to objectively assess the artists' sense of control and authorship over the creative process[21]. During the sessions, participants are encouraged to verbalize their thoughts, while we document their interactions, inner thought processes, and reported emotional feedback. The interviews aim to gain a better and deeper understanding of the artists' perceptions, experiences, and opinions regarding their collaboration with INDRA. Surveys are conducted before and after the painting sessions to capture any changes in the artists' attitudes toward AI in art.

5 Application Scenario

- (1) **AI in Art Education:** Traditional painting education often relies on in-person mentorship, or resources like books and

online tutorials, which can lack personalized and hands-on guidance. The National Art Education Association recognizes generative AI as both an opportunity and a challenge for visual art education. They emphasize the importance of balancing AI tools with traditional art forms to ensure that individual expression and human-created art are not overshadowed[17]. In an art classroom or during at-home learning, use of INDRA can help students develop their painting skills by providing real-time feedback and narrative suggestions based on their brushstrokes. For effective implementation, we propose demonstrating techniques such as rag wiping, glazing, or dry brush effects. Beginners will receive structured guidance to build their confidence and encourage experimentation.

- (2) **Add-on in a traditional painting studio:** For traditional studio artists, INDRA acts as a collaborator, adapting to their vision and enhancing their narrative. It analyzes brushstrokes and suggests relevant additions to inspire creativity without overshadowing their style. This dynamic partnership encourages artists to explore new possibilities, blending traditional and generative techniques. INDRA amplifies the artist's voice, blending traditional and generative techniques for new possibilities.
- (3) **Accessibility in Painting:** INDRA offers accessibility and inclusion for artists facing physical limitations, enabling

them to create on a canvas through a tablet interface, with INDRA replicating their movements. Its speech recognition feature allows users to verbally guide the robotic arm, empowering them to fully express their artistic vision in physical traditional arts. For individuals exploring creative therapy or rediscovering art after even with bodily limitations. Moreover, INDRA's assistant mode enhances accessibility by enabling users to paint indirectly through digital drawings on an iPad. This feature seamlessly translates their intentions into precise robotic movements, effectively bridging the gap between human creativity and AI assistance.

6 Conclusion and Future Work

This research highlights the importance of designing AI systems that enhance, rather than overshadow, human creativity. By fostering a collaborative environment, INDRA demonstrates how embodied AI can merge with traditional art forms to augment artistic agency and innovation, offering a new dimension to creative expression. In future work, we envision INDRA empowering artists with physical limitations to engage fully in large-scale painting by integrating technologies like eye tracking for intuitive control. Future developments include refining the generative pipeline and introducing features like a real-time "preview" mode, enabling users to view and adjust the AI's proposed actions. These enhancements aim to create a more inclusive, intuitive, and supportive creative environment, ensuring that all artists can fully realize their vision.

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